Jason Pereira

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EDUCATION

University of North Carolina at Chapel Hill | Chapel Hill, NC **May 2028**

B.S. in Computer Science, B.S. in Statistics and Analytics, Minor in Business | Honors Carolina, Dean's List

- **Relevant Coursework:** Data Structures & Algorithms, Systems Fundamentals, Discrete Math, Foundations of Programming, Linear Algebra, Computer Organization, Models of Language & Computation, 2D Computer Graphics
- **Clubs:** App Team Carolina (iOS Developer), HackNC Development Committee, CS + Social Good, Carolina Investment Group

EXPERIENCE

Amazon Web Services (AWS), *Solutions Architect Intern* | Arlington, VA **May 2026 - Aug 2026**

- Developed a serverless AI Ad Copy Generator with **4-agent** architecture on **Amazon Bedrock AgentCore**, generating **20+** platform-specific outputs in **<5 minutes** through RAG-backed Strategist, Copywriter, Compliance, and Critic agents
- Engineered an authenticated AWS stack with Cognito, API Gateway, Lambda, DynamoDB, S3, and OpenSearch Serverless, separating model inference from a **sub-300 ms** serving path and achieving **90%+** restricted-term detection in testing
- Co-led development of **5** reusable industry demo packs informed by interviews with 60+ Solutions Architects, scaling the initiative to **~2,000 SAs** and reducing demo preparation from **60+ mins to <5 minutes**; presented to leadership in NYC

Arthmetis, *AI Engineer Intern* | Charlotte, NC **Jun 2025 - Aug 2025**

- Built a hybrid retail recommendation engine in **Python, LangChain, FAISS, and GPT-4**, combining semantic search with customer demographics, purchase history, location, and seasonality to generate personalized product recommendations
- Trained an **XGBoost purchase-propensity** model on **50K+** customer-product interactions, incorporating behavioral, demographic, product, and vector-similarity features to improve top-3 recommendation precision by **25%**

UNC Charlotte, *Data Science Researcher* | Charlotte, NC **Feb 2024 - May 2026**

- Co-authored **2 data science studies** examining racial disparities, analyzing a 5,000-stop sample from **124K traffic-stop records** and statewide school discipline across **all 100 NC districts** using Python, CODAP, data visualization, and inference
- Published **NSF-supported findings to TEEM** and presented findings to educators at NCCTM State Conference, finding certain drivers were **2.4x more likely to be searched** and made up 56% of stops despite comprising 33% of Charlotte residents

PROJECTS

Case Closed | *Duke AI Hackathon, **1st Place**, Productivity & Enterprise Tools Track | Backed by 1789 Fund* **Nov 2025 - Present**

- Built an AI litigation workspace in **Python/Flask and GCP** that ingests case records and returns authority grounded in retrieved opinions, using Gemini on Vertex AI over the CourtListener API across **9M+ opinions**; 1st place in 220+ competitors
- Engineered citation-grounded retrieval with **Cloud Vector Search and CourtListener API**, requiring every generated citation to resolve to retrieved authority; added Clerk auth, Docker, and Cloud Run; now in early access with attorneys

StreetLab: Real-Time Autonomous-Driving Simulator | *Personal Project* | Rust, Tauri, React, Three.js, WebGPU **Aug 2026**

- Built a real-time autonomous-driving simulator that drives vehicles through real OpenStreetMap roads, using Rust/Tauri, React/Three.js/WebGPU, and **Python/FastAPI** to render and stream simulation data at **30–60 FPS**
- Developed realistic driving behavior, traffic-light and stop-sign handling, lane changes, and RT-DETR computer vision; maintained **1,100+ automated tests** and validated performance across **14,000+ simulation frames** on real-world roads

AWARDS & LEADERSHIP

- **1st Place, Anthropic Claude AI Hackathon (2025)**, built an interactive gamified stress-relief productivity app
- **Eagle Scout**, *Scouting America*, active since 2013, with 32 Merit Badges, Gold Eagle Palm. Led 181-hour Eagle Service Project
- **TEDx Speaker**, "The Velocity Paradox," *TEDxUNC* (Feb 2026); **Charlotte's 20 Under 20**, *The Dean's List*
- **4th Place, National Leadership Conference**, *FBLA*, Website Design; **2x North Carolina State Champion - 1st Place**

SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, Swift, SQL, HTML/CSS
Systems & Infrastructure: Unix/Linux, CUDA/GPU, distributed systems, Docker, AWS (Lambda, EC2, Bedrock), Git, CI/CD, REST APIs
Frameworks & Data: React, Next.js, Node.js, FastAPI, Flask, SwiftUI, PostgreSQL, MongoDB, PyTorch, RAG, FAISS, pgvector, XGBoost
Certifications: AWS Solutions Architect – Associate, AWS Machine Learning Engineer – Associate, Google Gen AI Leader